

Role of mechanical stress localizations on the radiation hardness of AlGa_N/Ga_N high electron mobility transistors

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Abstract

Multi-material, multi-layered systems such as AlGa_N/Ga_N high electron mobility transistors (HEMTs) contain residual mechanical stresses that arise from sharp contrasts in device geometry and materials parameters. These stresses, which can be either tensile or compressive, are difficult to detect and eliminate because of their highly localized nature. We propose that their high-stored internal energy makes potential sites for defect nucleation sites under radiation, particularly if their locations coincide with the electrically sensitive regions of a transistor. In this study, we validate this hypothesis with molecular dynamic simulation and experiments exposing both pristine and annealed HEMTS to 2.8 MeV Au⁺³ irradiation. Our unique annealing process uses mechanical momentum of electrons, also known as the electron wind force (EWF) to mitigate the residual stress at room temperature. High-resolution transmission electron microscopy and cathodoluminescence spectra reveal the reduction of point defects and dislocations near the two-dimensional electron gas region of EWF-treated devices compared to pristine devices. The EWF-treated HEMTs showed relatively higher resilience with approximately 10% less degradation of drain saturation current and ON-resistance and 5% less degradation of peak transconductance. Both mobility and carrier

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